DBMS DML Statements

Project Title :A System of Smart Power Grid Management and Energy Resource Allocation

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Insert Queries

1. INSERT-USER

```
mysql> INSERT INTO USER
    -> (name, email, password_hash, role, phone, status)
    -> VALUES
    -> ('Arjun Sharma', 'arjun@gmail.com', 'hash001', 'ADMIN', '9876543210', 'ACTIVE'),
    -> ('Meera Rao', 'meera@gmail.com', 'hash002', 'OPERATOR', '9876543211', 'ACTIVE'),
    -> ('Rahul Kumar', 'rahul@gmail.com', 'hash003', 'CONSUMER', '9876543212', 'ACTIVE'),
    -> ('Sneha Patel', 'sneha@gmail.com', 'hash004', 'CONSUMER', '9876543213', 'ACTIVE'),
    -> ('Vikram Singh', 'vikram@gmail.com', 'hash005', 'OPERATOR', '9876543214', 'ACTIVE');
Query OK, 5 rows affected (0.05 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

2. INSERT GRID_ZONE

```
mysql> INSERT INTO GRID_ZONE
    -> (zone_name, zone_type, priority_level, max_capacity_mw,
    ->  current_demand_mw, allocated_power_mw, status, reported_level)
    -> VALUES
    -> ('North Residential Zone', 'RESIDENTIAL', 2, 100.00, 65.00, 60.00, 'NORMAL', 'LOW'),
    -> ('Central Commercial Zone', 'COMMERCIAL', 3, 150.00, 120.00, 110.00, 'HIGH_DEMAND', 'MEDIUM'),
    -> ('East Industrial Zone', 'INDUSTRIAL', 4, 250.00, 210.00, 200.00, 'HIGH_DEMAND', 'HIGH'),
    -> ('City Hospital Zone', 'CRITICAL', 5, 80.00, 70.00, 70.00, 'NORMAL', 'LOW');
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

3. INSERT POWER_PLANT

```
mysql> INSERT INTO POWER_PLANT
    -> (plant_name, plant_type, location, capacity_mw,
    ->  current_generation_mw, status, commissioned_date)
    -> VALUES
    -> ('Green Solar Plant', 'SOLAR', 'Bangalore', 200.00, 150.00, 'ACTIVE', '2022-06-15'),
    -> ('Coastal Wind Plant', 'WIND', 'Mangalore', 180.00, 130.00, 'ACTIVE', '2021-09-10'),
    -> ('River Hydro Plant', 'HYDRO', 'Shimoga', 300.00, 250.00, 'ACTIVE', '2020-03-20'),
    -> ('Thermal Power Station', 'THERMAL', 'Raichur', 500.00, 400.00, 'ACTIVE', '2019-01-10');
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

4. INSERT CONSUMER

```
mysql> INSERT INTO CONSUMER
    -> (user_id, consumer_type, address, connection_date, status)
    -> VALUES
    -> (3, 'RESIDENTIAL', 'Whitefield, Bangalore', '2024-01-15', 'ACTIVE'),
    -> (4, 'COMMERCIAL', 'MG Road, Bangalore', '2024-02-10', 'ACTIVE');
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

5. INSERT METER

```
mysql> INSERT INTO METER
    -> (consumer_id, zone_id, meter_type, installation_date, status)
    -> VALUES
    -> (1, 1, 'SMART', '2024-01-20', 'ACTIVE'),
    -> (2, 2, 'SMART', '2024-02-15', 'ACTIVE');
Query OK, 2 rows affected (0.08 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

6. INSERT OUTAGE

```
mysql> INSERT INTO OUTAGE
    -> (zone_id, reported_by, start_time, end_time, reason, severity, status)
    -> VALUES
    -> (1, 1, '2026-09-01 10:00:00', '2026-09-01 12:00:00',
    ->  'Transformer Failure', 'HIGH', 'RESOLVED'),
    ->
    -> (2, 2, '2026-09-05 14:00:00', '2026-09-05 15:30:00',
    ->  'Transmission Fault', 'MEDIUM', 'RESOLVED'),
    ->
    -> (3, 2, '2026-09-10 09:00:00', NULL,
    ->  'Equipment Failure', 'CRITICAL', 'ACTIVE');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

7. INSERT ENERGY_RESOURCE

```
mysql> INSERT INTO ENERGY_RESOURCE
    -> (resource_name, resource_type, capacity_mw, available_power_mw,
    ->  location, status)
    -> VALUES
    -> ('Battery Storage A', 'BATTERY', 100.00, 80.00, 'North Zone', 'AVAILABLE'),
    -> ('Backup Generator B', 'GENERATOR', 150.00, 120.00, 'Central Zone', 'AVAILABLE'),
    -> ('Emergency Reserve C', 'RESERVE', 200.00, 200.00, 'East Zone', 'AVAILABLE'),
    -> ('Grid Storage D', 'STORAGE', 120.00, 90.00, 'Central Zone', 'IN_USE');
Query OK, 4 rows affected (0.06 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

8. INSERT ENERGY_ALLOCATION

```
mysql> INSERT INTO ENERGY_ALLOCATION
    -> (plant_id, zone_id, allocated_power_mw,
    ->  allocation_time, priority_level, status, notes)
    -> VALUES
    -> (1, 1, 50.00, '2026-09-20 10:00:00', 2, 'COMPLETED',
    ->  'Residential demand allocation'),
    ->
    -> (2, 2, 80.00, '2026-09-20 11:00:00', 3, 'APPROVED',
    ->  'Commercial demand allocation'),
    ->
    -> (3, 3, 120.00, '2026-09-20 12:00:00', 4, 'COMPLETED',
    ->  'Industrial power requirement'),
    ->
    -> (4, 4, 60.00, '2026-09-20 13:00:00', 5, 'PENDING',
    ->  'Critical hospital requirement');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

9. INSERT RESOURCE_ALLOCATION

```
mysql> INSERT INTO RESOURCE_ALLOCATION
    -> (allocation_id, resource_id, power_contributed_mw)
    -> VALUES
    -> (1, 1, 20.00),
    -> (2, 2, 40.00),
    -> (3, 3, 60.00),
    -> (4, 4, 30.00);
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

10. INSERT ALLOCATION_LOG

```
mysql> INSERT INTO ALLOCATION_LOG
    -> (allocation_id, changed_by, old_power_mw,
    ->  new_power_mw, reason)
    -> VALUES
    -> (1, 2, 40.00, 50.00, 'Increased residential demand'),
    ->
    -> (2, 2, 70.00, 80.00, 'Commercial peak demand'),
    ->
    -> (3, 5, 100.00, 120.00, 'Industrial requirement');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

11. INSERT MAINTENANCE

```
mysql> INSERT INTO MAINTENANCE
    -> (plant_id, resource_id, maintenance_type,
    ->  scheduled_date, completion_date, status,
    ->  performed_by, description)
    -> VALUES
    -> (1, NULL, 'ROUTINE', '2026-10-01', NULL,
    ->  'SCHEDULED', 2, 'Solar panel inspection'),
    ->
    -> (NULL, 2, 'REPAIR', '2026-09-25', '2026-09-27',
    ->  'COMPLETED', 5, 'Generator maintenance'),
    ->
    -> (3, NULL, 'EMERGENCY', '2026-09-20', '2026-09-21',
    ->  'COMPLETED', 2, 'Hydro turbine repair');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

UPDATE queries

Update 1 — Change plant generation

```
mysql> UPDATE POWER_PLANT
    -> SET current_generation_mw = 175.00
    -> WHERE plant_id = 1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
    -> FROM POWER_PLANT
    -> WHERE plant_id = 1;
+----------+------------------+------------+-----------+-------------+----------------------+--------+-------------------+
| plant_id | plant_name       | plant_type | location  | capacity_mw | current_generation_mw | status | commissioned_date |
+----------+------------------+------------+-----------+-------------+----------------------+--------+-------------------+
|        1 | Green Solar Plant | SOLAR      | Bangalore |      200.00 |               175.00 | ACTIVE | 2022-06-15        |
+----------+------------------+------------+-----------+-------------+----------------------+--------+-------------------+
1 row in set (0.00 sec)
```

Update 2 — Change zone status

```
mysql> UPDATE GRID_ZONE
    -> SET status = 'HIGH_DEMAND',
    ->     reported_level = 'HIGH'
    -> WHERE zone_id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Update 3 — Update allocation status

```
mysql> UPDATE ENERGY_ALLOCATION
    -> SET status = 'COMPLETED'
    -> WHERE allocation_id = 4;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

DELETE query

```
mysql> INSERT INTO ENERGY_RESOURCE
    -> (resource_name, resource_type, capacity_mw,
    ->  available_power_mw, location, status)
    -> VALUES
    -> ('Temporary Generator', 'GENERATOR', 50.00, 50.00,
    ->  'Test Location', 'AVAILABLE');
Query OK, 1 row affected (0.06 sec)

mysql> SELECT *
    -> FROM ENERGY_RESOURCE
    -> WHERE resource_name = 'Temporary Generator';
+-------------+--------------------+---------------+-------------+--------------------+---------------+-----------+
| resource_id | resource_name      | resource_type | capacity_mw | available_power_mw | location      | status    |
+-------------+--------------------+---------------+-------------+--------------------+---------------+-----------+
|           5 | Temporary Generator | GENERATOR     |       50.00 |              50.00 | Test Location | AVAILABLE |
+-------------+--------------------+---------------+-------------+--------------------+---------------+-----------+
1 row in set (0.00 sec)

mysql> DELETE FROM ENERGY_RESOURCE
    -> WHERE resource_name = 'Temporary Generator';
Query OK, 1 row affected (0.06 sec)

mysql> SELECT *
    -> FROM ENERGY_RESOURCE
    -> WHERE resource_name = 'Temporary Generator';
Empty set (0.00 sec)
```

SELECT — Basic queries

```
mysql> SELECT *
    -> FROM POWER_PLANT;
+----------+-----------------------+------------+-----------+------------+----------------------+--------+------------------+
| plant_id | plant_name            | plant_type | location  | capacity_mw | current_generation_mw | status | commissioned_date |
+----------+-----------------------+------------+-----------+------------+----------------------+--------+------------------+
|        1 | Green Solar Plant     | SOLAR      | Bangalore |      200.00 |               175.00 | ACTIVE | 2022-06-15        |
|        2 | Coastal Wind Plant    | WIND       | Mangalore |      180.00 |               130.00 | ACTIVE | 2021-09-10        |
|        3 | River Hydro Plant     | HYDRO      | Shimoga   |      300.00 |               250.00 | ACTIVE | 2020-03-20        |
|        4 | Thermal Power Station | THERMAL    | Raichur   |      500.00 |               400.00 | ACTIVE | 2019-01-10        |
+----------+-----------------------+------------+-----------+------------+----------------------+--------+------------------+
4 rows in set (0.00 sec)

mysql> SELECT plant_name, plant_type, capacity_mw
    -> FROM POWER_PLANT
    -> WHERE status = 'ACTIVE';
+-----------------------+------------+-------------+
| plant_name            | plant_type | capacity_mw |
+-----------------------+------------+-------------+
| Green Solar Plant     | SOLAR      |      200.00 |
| Coastal Wind Plant    | WIND       |      180.00 |
| River Hydro Plant     | HYDRO      |      300.00 |
| Thermal Power Station | THERMAL    |      500.00 |
+-----------------------+------------+-------------+
4 rows in set (0.00 sec)

mysql> SELECT plant_name, capacity_mw
    -> FROM POWER_PLANT
    -> WHERE capacity_mw > 200
    -> ORDER BY capacity_mw DESC;
+-----------------------+-------------+
| plant_name            | capacity_mw |
+-----------------------+-------------+
| Thermal Power Station |      500.00 |
| River Hydro Plant     |      300.00 |
+-----------------------+-------------+
2 rows in set (0.03 sec)
```

JOIN

```
mysql> SELECT
    ->     ea.allocation_id,
    ->     pp.plant_name,
    ->     gz.zone_name,
    ->     ea.allocated_power_mw,
    ->     ea.priority_level,
    ->     ea.status
    -> FROM ENERGY_ALLOCATION ea
    -> JOIN POWER_PLANT pp
    ->     ON ea.plant_id = pp.plant_id
    -> JOIN GRID_ZONE gz
    ->     ON ea.zone_id = gz.zone_id;
+---------------+-----------------------+-------------------------+--------------------+----------------+-----------+
| allocation_id | plant_name            | zone_name               | allocated_power_mw | priority_level | status    |
+---------------+-----------------------+-------------------------+--------------------+----------------+-----------+
|             1 | Green Solar Plant     | North Residential Zone  |              50.00 |              2 | COMPLETED |
|             2 | Coastal Wind Plant    | Central Commercial Zone |              80.00 |              3 | APPROVED  |
|             3 | River Hydro Plant     | East Industrial Zone    |             120.00 |              4 | COMPLETED |
|             4 | Thermal Power Station | City Hospital Zone      |              60.00 |              5 | COMPLETED |
+---------------+-----------------------+-------------------------+--------------------+----------------+-----------+
4 rows in set (0.00 sec)
```

JOIN — Consumer + User

```
mysql> SELECT
    ->     c.consumer_id,
    ->     u.name,
    ->     u.email,
    ->     c.consumer_type,
    ->     c.address,
    ->     c.status
    -> FROM CONSUMER c
    -> JOIN USER u
    ->     ON c.user_id = u.user_id;
+-------------+-------------+-----------------+---------------+-----------------------+---------+
| consumer_id | name        | email           | consumer_type | address               | status  |
+-------------+-------------+-----------------+---------------+-----------------------+---------+
|           1 | Rahul Kumar | rahul@gmail.com | RESIDENTIAL   | Whitefield, Bangalore | ACTIVE  |
|           2 | Sneha Patel | sneha@gmail.com | COMMERCIAL    | MG Road, Bangalore    | ACTIVE  |
+-------------+-------------+-----------------+---------------+-----------------------+---------+
2 rows in set (0.00 sec)
```

JOIN — Meter + Consumer + Zone

```
mysql> SELECT
    ->     m.meter_id,
    ->     u.name AS consumer_name,
    ->     gz.zone_name,
    ->     m.meter_type,
    ->     m.status
    -> FROM METER m
    -> JOIN CONSUMER c
    ->     ON m.consumer_id = c.consumer_id
    -> JOIN USER u
    ->     ON c.user_id = u.user_id
    -> JOIN GRID_ZONE gz
    ->     ON m.zone_id = gz.zone_id;
+----------+---------------+------------------------+------------+--------+
| meter_id | consumer_name | zone_name              | meter_type | status |
+----------+---------------+------------------------+------------+--------+
|        1 | Rahul Kumar   | North Residential Zone | SMART      | ACTIVE |
|        2 | Sneha Patel   | Central Commercial Zone | SMART      | ACTIVE |
+----------+---------------+------------------------+------------+--------+
2 rows in set (0.00 sec)
```

GROUP BY

```
mysql> SELECT
    ->     pp.plant_name,
    ->     SUM(ea.allocated_power_mw) AS total_allocated_power
    -> FROM ENERGY_ALLOCATION ea
    -> JOIN POWER_PLANT pp
    ->     ON ea.plant_id = pp.plant_id
    -> GROUP BY pp.plant_name;
+----------------------+-----------------------+
| plant_name           | total_allocated_power |
+----------------------+-----------------------+
| Green Solar Plant    |                 50.00 |
| Coastal Wind Plant   |                 80.00 |
| River Hydro Plant    |                120.00 |
| Thermal Power Station |                60.00 |
+----------------------+-----------------------+
4 rows in set (0.00 sec)
```

GROUP BY + HAVING

```
mysql> SELECT
    ->     pp.plant_name,
    ->     SUM(ea.allocated_power_mw) AS total_allocated_power
    -> FROM ENERGY_ALLOCATION ea
    -> JOIN POWER_PLANT pp
    ->     ON ea.plant_id = pp.plant_id
    -> GROUP BY pp.plant_name
    -> HAVING SUM(ea.allocated_power_mw) > 100;
+------------------+-----------------------+
| plant_name       | total_allocated_power |
+------------------+-----------------------+
| River Hydro Plant |                120.00 |
+------------------+-----------------------+
1 row in set (0.00 sec)
```

COUNT + GROUP BY

```
mysql> SELECT
    ->     status,
    ->     COUNT(*) AS number_of_allocations
    -> FROM ENERGY_ALLOCATION
    -> GROUP BY status;
+-----------+-----------------------+
| status    | number_of_allocations |
+-----------+-----------------------+
| COMPLETED |                     3 |
| APPROVED  |                     1 |
+-----------+-----------------------+
2 rows in set (0.00 sec)
```

AVG

```
mysql> SELECT
    ->     AVG(current_generation_mw) AS average_generation
    -> FROM POWER_PLANT;
+--------------------+
| average_generation |
+--------------------+
|         238.750000 |
+--------------------+
1 row in set (0.02 sec)
```

MAX and MIN

```
mysql> SELECT
    ->     MAX(capacity_mw) AS maximum_capacity,
    ->     MIN(capacity_mw) AS minimum_capacity
    -> FROM POWER_PLANT;
+------------------+------------------+
| maximum_capacity | minimum_capacity |
+------------------+------------------+
|           500.00 |           180.00 |
+------------------+------------------+
1 row in set (0.06 sec)
```

ORDER BY

```
mysql> SELECT
    ->     plant_name,
    ->     capacity_mw,
    ->     current_generation_mw
    -> FROM POWER_PLANT
    -> ORDER BY capacity_mw DESC;
+----------------------+-------------+----------------------+
| plant_name           | capacity_mw | current_generation_mw |
+----------------------+-------------+----------------------+
| Thermal Power Station |      500.00 |                400.00 |
| River Hydro Plant     |      300.00 |                250.00 |
| Green Solar Plant     |      200.00 |                175.00 |
| Coastal Wind Plant    |      180.00 |                130.00 |
+----------------------+-------------+----------------------+
4 rows in set (0.00 sec)
```

SUBQUERY

Find plants whose capacity is greater than the average plant capacity

```
mysql> SELECT
    ->     plant_name,
    ->     capacity_mw
    -> FROM POWER_PLANT
    -> WHERE capacity_mw >
    -> (
    ->     SELECT AVG(capacity_mw)
    ->     FROM POWER_PLANT
    -> );
+----------------------+-------------+
| plant_name           | capacity_mw |
+----------------------+-------------+
| River Hydro Plant     |      300.00 |
| Thermal Power Station |      500.00 |
+----------------------+-------------+
2 rows in set (0.00 sec)
```

Find zones whose current demand is greater than the average demand:

```
mysql> SELECT
    ->     zone_name,
    ->     current_demand_mw
    -> FROM GRID_ZONE
    -> WHERE current_demand_mw >
    -> (
    ->     SELECT AVG(current_demand_mw)
    ->     FROM GRID_ZONE
    -> );
+-------------------------+--------------------+
| zone_name               | current_demand_mw  |
+-------------------------+--------------------+
| Central Commercial Zone |             120.00 |
| East Industrial Zone    |             210.00 |
+-------------------------+--------------------+
2 rows in set (0.00 sec)
```

A good complex query

```
mysql> SELECT
    ->     gz.zone_name,
    ->     gz.zone_type,
    ->     COUNT(ea.allocation_id) AS number_of_allocations,
    ->     SUM(ea.allocated_power_mw) AS total_power_allocated,
    ->     AVG(ea.allocated_power_mw) AS average_power_allocated
    -> FROM GRID_ZONE gz
    -> JOIN ENERGY_ALLOCATION ea
    ->     ON gz.zone_id = ea.zone_id
    -> GROUP BY
    ->     gz.zone_id,
    ->     gz.zone_name,
    ->     gz.zone_type
    -> HAVING SUM(ea.allocated_power_mw) > 50
    -> ORDER BY total_power_allocated DESC;
+-------------------------+------------+----------------------+----------------------+--------------------------+
| zone_name               | zone_type  | number_of_allocations | total_power_allocated | average_power_allocated |
+-------------------------+------------+----------------------+----------------------+--------------------------+
| East Industrial Zone    | INDUSTRIAL |                    1 |               120.00 |               120.000000 |
| Central Commercial Zone | COMMERCIAL |                    1 |                80.00 |                80.000000 |
| City Hospital Zone      | CRITICAL   |                    1 |                60.00 |                60.000000 |
+-------------------------+------------+----------------------+----------------------+--------------------------+
3 rows in set (0.06 sec)
```

ADVANCED DATABASE FEATURES

1.Views

```
mysql> CREATE VIEW vw_power_allocation_summary AS
    -> SELECT
    ->     ea.allocation_id,
    ->     pp.plant_name,
    ->     gz.zone_name,
    ->     ea.allocated_power_mw,
    ->     ea.allocation_time,
    ->     ea.priority_level,
    ->     ea.status
    -> FROM ENERGY_ALLOCATION ea
    -> JOIN POWER_PLANT pp
    ->     ON ea.plant_id = pp.plant_id
    -> JOIN GRID_ZONE gz
    ->     ON ea.zone_id = gz.zone_id;
ERROR 1050 (42S01): Table 'vw_power_allocation_summary' already exists
mysql> SELECT * FROM vw_power_allocation_summary;
+---------------+---------------------+-------------------------+-------------------+---------------------+----------------+-----------+
| allocation_id | plant_name          | zone_name               | allocated_power_mw | allocation_time     | priority_level | status    |
+---------------+---------------------+-------------------------+-------------------+---------------------+----------------+-----------+
|             1 | Green Solar Plant   | North Residential Zone  |             50.00 | 2026-09-20 10:00:00 |              2 | COMPLETED |
|             2 | Coastal Wind Plant  | Central Commercial Zone |             80.00 | 2026-09-20 11:00:00 |              3 | APPROVED  |
|             3 | River Hydro Plant   | East Industrial Zone    |            120.00 | 2026-09-20 12:00:00 |              4 | COMPLETED |
|             4 | Thermal Power Station | City Hospital Zone    |             60.00 | 2026-09-20 13:00:00 |              5 | COMPLETED |
+---------------+---------------------+-------------------------+-------------------+---------------------+----------------+-----------+
4 rows in set (0.00 sec)
```

2. Transaction using START TRANSACTION and COMMIT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE ENERGY_ALLOCATION
    -> SET allocated_power_mw = allocated_power_mw + 10
    -> WHERE allocation_id = 1;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> UPDATE GRID_ZONE
    -> SET allocated_power_mw = allocated_power_mw + 10
    -> WHERE zone_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT *
    -> FROM ENERGY_ALLOCATION
    -> WHERE allocation_id = 1;
+---------------+----------+---------+-------------------+---------------------+----------------+-----------+-------------------------------+
| allocation_id | plant_id | zone_id | allocated_power_mw | allocation_time     | priority_level | status    | notes                         |
+---------------+----------+---------+-------------------+---------------------+----------------+-----------+-------------------------------+
|             1 |        1 |       1 |             60.00 | 2026-09-20 10:00:00 |              2 | COMPLETED | Residential demand allocation |
+---------------+----------+---------+-------------------+---------------------+----------------+-----------+-------------------------------+
1 row in set (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM GRID_ZONE
    -> WHERE zone_id = 1;
+---------+-----------------------+-------------+----------------+-----------------+-------------------+--------------------+-------------+----------------+
| zone_id | zone_name             | zone_type   | priority_level | max_capacity_mw | current_demand_mw | allocated_power_mw | status      | reported_level |
+---------+-----------------------+-------------+----------------+-----------------+-------------------+--------------------+-------------+----------------+
|       1 | North Residential Zone | RESIDENTIAL |              2 |          100.00 |             65.00 |              70.00 | HIGH_DEMAND | HIGH           |
+---------+-----------------------+-------------+----------------+-----------------+-------------------+--------------------+-------------+----------------+
1 row in set (0.00 sec)
```

3. Transaction rollback demonstration

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE ENERGY_ALLOCATION
    -> SET allocated_power_mw = 999
    -> WHERE allocation_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT allocated_power_mw
    -> FROM ENERGY_ALLOCATION
    -> WHERE allocation_id = 1;
+--------------------+
| allocated_power_mw |
+--------------------+
|             999.00 |
+--------------------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT allocated_power_mw
    -> FROM ENERGY_ALLOCATION
    -> WHERE allocation_id = 1;
+--------------------+
| allocated_power_mw |
+--------------------+
|              60.00 |
+--------------------+
1 row in set (0.00 sec)
```

4. Trigger automatically logging changes to ENERGY_ALLOCATION

```
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER trg_allocation_update_log
    -> AFTER UPDATE ON ENERGY_ALLOCATION
    -> FOR EACH ROW
    -> BEGIN
    ->
    ->     IF OLD.allocated_power_mw <> NEW.allocated_power_mw THEN
    ->
    ->         INSERT INTO ALLOCATION_LOG
    ->         (
    ->             allocation_id,
    ->             changed_by,
    ->             old_power_mw,
    ->             new_power_mw,
    ->             changed_at,
    ->             reason
    ->         )
    ->         VALUES
    ->         (
    ->             NEW.allocation_id,
    ->             2,
    ->             OLD.allocated_power_mw,
    ->             NEW.allocated_power_mw,
    ->             CURRENT_TIMESTAMP,
    ->             'Power allocation updated'
    ->         );
    ->
    ->     END IF;
    ->
    -> END //
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT user_id, name, role
    -> FROM USER;
+---------+--------------+----------+
| user_id | name         | role     |
+---------+--------------+----------+
|       1 | Arjun Sharma | ADMIN    |
|       2 | Meera Rao    | OPERATOR |
|       3 | Rahul Kumar  | CONSUMER |
|       4 | Sneha Patel  | CONSUMER |
|       5 | Vikram Singh | OPERATOR |
+---------+--------------+----------+
5 rows in set (0.01 sec)

mysql> UPDATE ENERGY_ALLOCATION
    -> SET allocated_power_mw = allocated_power_mw + 20
    -> WHERE allocation_id = 1;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT *
    -> FROM ALLOCATION_LOG
    -> WHERE allocation_id = 1
    -> ORDER BY changed_at DESC;
+--------+---------------+------------+--------------+--------------+---------------------+-----------------------------+
| log_id | allocation_id | changed_by | old_power_mw | new_power_mw | changed_at          | reason                      |
+--------+---------------+------------+--------------+--------------+---------------------+-----------------------------+
|      4 |             1 |          2 |        60.00 |        80.00 | 2026-10-02 22:52:31 | Power allocation updated    |
|      1 |             1 |          2 |        40.00 |        50.00 | 2026-09-29 09:26:05 | Increased residential demand |
+--------+---------------+------------+--------------+--------------+---------------------+-----------------------------+
2 rows in set (0.00 sec)
```

5. Execution of stored procedure for retrieving zone allocations

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetZoneAllocations(IN p_zone_id INT)
    -> BEGIN
    ->
    ->     SELECT
    ->         ea.allocation_id,
    ->         pp.plant_name,
    ->         gz.zone_name,
    ->         ea.allocated_power_mw,
    ->         ea.priority_level,
    ->         ea.status,
    ->         ea.allocation_time
    ->     FROM ENERGY_ALLOCATION ea
    ->
    ->     JOIN POWER_PLANT pp
    ->         ON ea.plant_id = pp.plant_id
    ->
    ->     JOIN GRID_ZONE gz
    ->         ON ea.zone_id = gz.zone_id
    ->
    ->     WHERE ea.zone_id = p_zone_id
    ->
    ->     ORDER BY ea.allocation_time DESC;
    ->
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetZoneAllocations(1);
+---------------+------------------+-----------------------+--------------------+----------------+-----------+---------------------+
| allocation_id | plant_name       | zone_name             | allocated_power_mw | priority_level | status    | allocation_time     |
+---------------+------------------+-----------------------+--------------------+----------------+-----------+---------------------+
|             1 | Green Solar Plant | North Residential Zone |              80.00 |              2 | COMPLETED | 2026-09-20 10:00:00 |
+---------------+------------------+-----------------------+--------------------+----------------+-----------+---------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

6. Execution of stored function to calculate power plant utilization

```
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION CalculatePlantUtilization(p_plant_id INT)
    -> RETURNS DECIMAL(5,2)
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE utilization DECIMAL(5,2);
    ->
    ->     SELECT
    ->         (current_generation_mw / capacity_mw) * 100
    ->     INTO utilization
    ->     FROM POWER_PLANT
    ->     WHERE plant_id = p_plant_id;
    ->
    ->     RETURN utilization;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> SELECT
    ->     plant_id,
    ->     plant_name,
    ->     capacity_mw,
    ->     current_generation_mw,
    ->     CalculatePlantUtilization(plant_id) AS utilization_percentage
    -> FROM POWER_PLANT;
+----------+----------------------+-------------+----------------------+------------------------+
| plant_id | plant_name           | capacity_mw | current_generation_mw | utilization_percentage |
+----------+----------------------+-------------+----------------------+------------------------+
|        1 | Green Solar Plant    |      200.00 |               175.00 |                  87.50 |
|        2 | Coastal Wind Plant   |      180.00 |               130.00 |                  72.22 |
|        3 | River Hydro Plant    |      300.00 |               250.00 |                  83.33 |
|        4 | Thermal Power Station |      500.00 |               400.00 |                  80.00 |
+----------+----------------------+-------------+----------------------+------------------------+
4 rows in set (0.01 sec)
```

7. Full-Text Search

Full-text search for energy resources

```
mysql> ALTER TABLE ENERGY_RESOURCE
    -> ADD FULLTEXT INDEX ft_resource_search
    -> (resource_name, location);
Query OK, 0 rows affected, 1 warning (0.20 sec)
Records: 0  Duplicates: 0  Warnings: 1

mysql> SHOW INDEX FROM ENERGY_RESOURCE;
+-----------------+------------+---------------------+--------------+---------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table           | Non_unique | Key_name            | Seq_in_index | Column_name   | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----------------+------------+---------------------+--------------+---------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| energy_resource |          0 | PRIMARY             |            1 | resource_id   | A         |           4 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| energy_resource |          1 | ft_resource_search |            1 | resource_name | NULL      |           4 |     NULL |   NULL |      | FULLTEXT   |         |               | YES     | NULL       |
| energy_resource |          1 | ft_resource_search |            2 | location      | NULL      |           4 |     NULL |   NULL | YES  | FULLTEXT   |         |               | YES     | NULL       |
+-----------------+------------+---------------------+--------------+---------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
3 rows in set (0.03 sec)

mysql> SELECT
    ->     resource_id,
    ->     resource_name,
    ->     resource_type,
    ->     capacity_mw,
    ->     available_power_mw,
    ->     location,
    ->     status
    -> FROM ENERGY_RESOURCE
    -> WHERE MATCH(resource_name, location)
    -> AGAINST('storage' IN NATURAL LANGUAGE MODE);
+-------------+------------------+---------------+-------------+------------------+-------------+-----------+
| resource_id | resource_name    | resource_type | capacity_mw | available_power_mw | location    | status    |
+-------------+------------------+---------------+-------------+------------------+-------------+-----------+
|           1 | Battery Storage A | BATTERY       |      100.00 |             80.00 | North Zone  | AVAILABLE |
|           4 | Grid Storage D    | STORAGE       |      120.00 |             90.00 | Central Zone | IN_USE    |
+-------------+------------------+---------------+-------------+------------------+-------------+-----------+
2 rows in set (0.01 sec)
```